Digital Voltmeter (Without Arduino)

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***Abstract*—The given project is the design and implementation of a digital voltmeter without the use of an Arduino or microcontroller. The main objective is to convert an analog input voltage into a digital value and display the corresponding voltage level on a 7-segment display. The system is fully implemented by hardware circuits and, therefore, it is an easy and economical method of learning about analog-to-digital conversion and digital display systems. This analog input voltage is then fed to a 3-bit Flash Analog-to-Digital Converter (ADC), which compares the analog input voltage against several fixed reference voltages with comparators. The outputs at the comparators are then sent through a 74LS148 priority encoder to produce a 3-bit binary value. Because the priority encoder itself generates active-LOW, the decoder is inverted with some NOT gates and then fed with the inverted signals into a BCD-to-7-segment decoder. The corresponding signals are then generated by the decoder to a common-anode 7-segment display, enabling the level of voltage measured to be represented as a digital value. The project shows how Flash ADCs, priority encoders, logic gates, and 7-segment display systems can be practically applied using hardware components alone.**

***Index Terms*—Digital Voltmeter, Flash ADC, analog to digital conversion, priority encoder, 74LS148, NOT gate, BCD to 7 segment decoder, 7 segment display**

I. Implementation

A. Working Principle

The digital voltmeter was designed by employing a 3-bit Flash Analog to Digital converter, a priority encoder, NOT gates, a BCD to 7 segment decoder, and a common anode 7 segment display. The full system transforms an analog value of the input voltage into a digital value, which can be viewed on the display. Since the entire conversion and display process was done with hardware circuits, no Arduino, microcontroller, or programming was needed.

The implementation was started by designing a 3-bit Flash ADC. To differentiate between the voltage levels, a 3 bit Flash ADC needs 7 comparators: $2^3 - 1 = 7$. The necessary reference voltages for the comparators were obtained by an 8-resistor voltage-divider network. The comparator part was done using LM324 operational amplifiers (Op-Amps) [1]. The analog input voltage, V_{in}, is applied to both comparator inputs, and the "resistor ladder" supplies fixed reference voltages. Each comparator thus only provides a logical signal as to whether the corresponding input voltage is larger or smaller than its reference voltage.

The outputs of the seven comparators were tied to a priority encoder – 74LS148, which encodes the outputs of the comparators to 3-bit binary output [2]. The outputs of the 74LS148, however, are active-LOW while the subsequent BCD-to-7-segment decoder requires active-HIGH [2], [3]. So NOT gates were added using 74LS04 to invert the three output signals between the priority encoder and the decoder [4].

The outputs of the NOT gates were then fed to the BCD-to-7-segment decoder. The binary input is converted by the decoder into the appropriate signals needed to turn on the 7-segment common-anode display segments [3]. Current limitation resistors between the outputs of the decoder and the display were used to reduce the current fed to each segment of the display.

Finally, the completed circuit had a variable input voltage applied to it. The binary output of the priority encoder was a function of the comparator output, which varied as the input voltage varied. The digital value coming out was shown on the 7 segment display after the inverted and, decoded value.

B. Circuit Components

Major components used for the implementation are seven comparators for a 3 bit Flash ADC, a voltage-divider ladder formed with 8 resistors, a 74LS148 priority encoder, NOT gates, BCD to 7 segment display, a common anode 7 segment display, and current limiting resistors. The analog to digital conversion and digital display functions are all accomplished by these components.

C. Circuit Diagram

The entire circuit diagram shows the relationship between the resistor ladder, the seven comparators, the 74LS148 priority encoder, the NOT gates, the BCD-to-7-segment decoder, and the common-anode 7-segment display. The comparators are connected to the analog input voltage, and the voltages

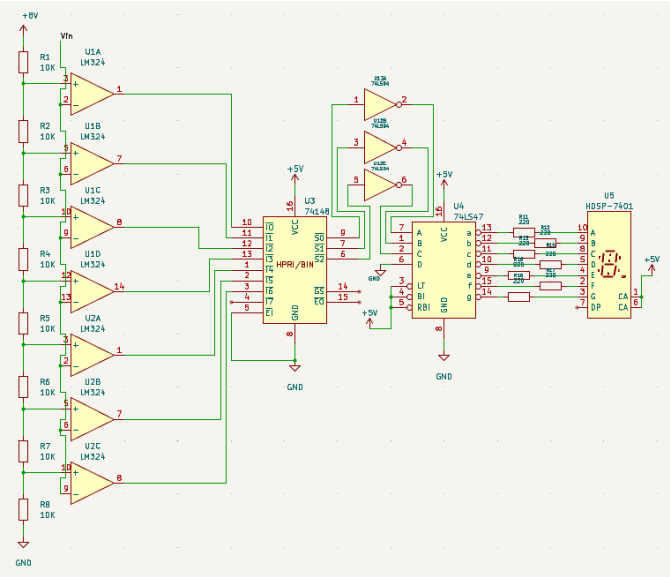


Fig. 1. Digital Voltmeter

necessary are provided by the resistor ladder. Outputs of the resulting comparators are fed through the priority encoder and NOT gates to the decoder and display.

D. Circuit Setup

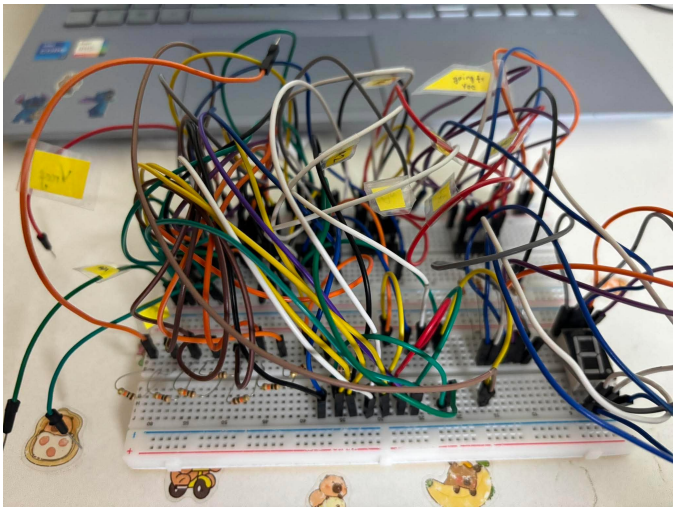


Fig. 2. Circuit Setup

E. Testing and Verification

Various levels of variable input voltage were used to test the completed circuit. At every input level, the comparator output was produced based on the respective reference voltages and the 74LS148 priority encoder created the correct 3-bit binary number. The digital value that was inverted and decoded was displayed on the 7-segment display. The work of both the Flash ADC and the display unit was checked by testing various input voltage levels.

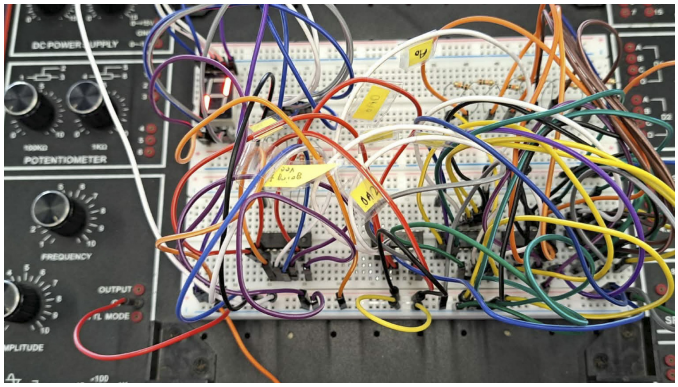


Fig. 3. Circuit Setup (continued)

F. Challenges and Solutions

An implementation problem was the difference in the logic polarity of the 74LS148 priority encoder and the BCD to 7-segment decoder. The priority encoder outputs are active-Low, the decoder inputs are active-High. The problem with this incompatibility was solved by adding NOT gates at the three outputs of the priority encoder [4]. The inverted signals could then be correctly interpreted by the decoder.

Another key aspect of the implementation was generation of suitable reference voltage to all seven comparators. This was rectified by making a voltage-divider ladder with 8 resistors. The resistor ladder supplies the fixed levels for the Flash ADC to compare to the variable levels.

II. Conclusion

A. Summary

The project successfully implemented the design of a digital voltmeter without using any microcontroller or Arduino. An analog input voltage was converted into a digital value by employing a 3-bit Flash ADC composed of seven comparators and an eight-resistor voltage-divider network. The output of the comparators was fed to a 74LS148 priority encoder, which was then inverted using NOT gates to give the correct logic levels to the BCD to 7-segment decoder. Finally, the output from the decoder was displayed on a common-anode 7-segment display.

The project was able to give hands on experience in analog to digital conversion, comparator circuit, priority encoders, logic gates, decoders, and digital displays. Different input voltages were tried out to ensure the functionality of the ADC and display unit.

B. Suggestions for Future Improvements

The system could be expanded and improved in the future to achieve greater resolution of measurement by implementing a more bits ADCs and additional reference levels. It may also be possible to add additional 7-segment displays to increase the accuracy of the voltage value, to the extent of adding decimal values. Other features like better measurement accuracy, auto range selection and signal conditioning can be added to further improve the functionality of the digital voltmeter.

REFERENCES

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